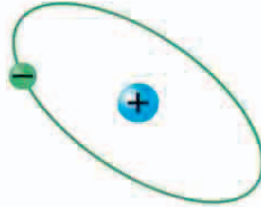


**Electron shell**

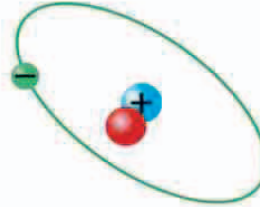
The electrons move around the nucleus, arranged in shells. Shells have a fixed number of spaces for electrons. In most cases, when one shell becomes full, another begins farther away from the nucleus.

**Isotopes**

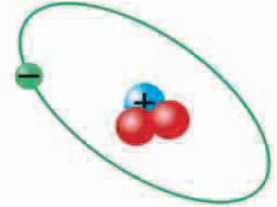
Atoms occur in different forms. While an element's atomic nucleus always has a certain number of protons, many contain different numbers of neutrons. These alternative versions of the atom are called isotopes. Atoms of different isotopes have varying weights.

**△ Hydrogen**

The main isotope of hydrogen has no neutrons in its nucleus.

**△ Deuterium**

With one extra neutron, this atom weighs twice as much as the main hydrogen isotope.

**△ Tritium**

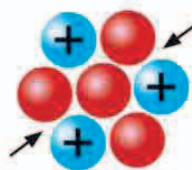
This hydrogen isotope is three times heavier than the main hydrogen isotope.

**REAL WORLD****Radiocarbon dating**

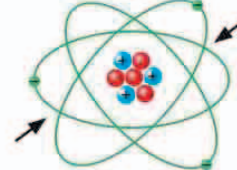
Scientists use the carbon-14 isotope to measure the age of ancient artifacts that are made from organic materials, such as wood or cotton. When new, the cotton wrapping of this mummy had a certain amount of carbon-14 in it. The isotope breaks down at a slow but fixed rate, and the amount left in the wrapping can tell scientists how old it is.

**Atomic forces**

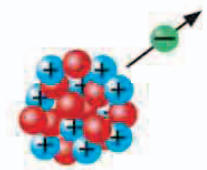
There are three forces at work inside atoms. The first type is a strong force that pulls the particles in the nucleus together. The second type occurs when the electrons are bonded to the atom by an electromagnetic force, which also acts over a much larger distance outside of the atom. The third type is a weak force that is involved in radioactivity, pushing particles out of the nucleus.

**△ Strong force**

This is the strongest force in nature, but it acts only over tiny distances.

**△ Electromagnetic force**

This force is involved in light and electricity, and holds atoms together.

**△ Weak force**

This force causes radioactive decay in atoms.

**Electron**

The electron has a negative charge that is equal and opposite to that of the proton. However, the mass of an electron is just a tiny fraction of a proton's mass.